Day 102 | Unit 07 | Teacher plan

Understand and Use Percent

Solve Percent Problems with Any Unknown

Standards: 6.RP.A.3c

Learning target: I can solve Percent Problems with Any Unknown, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

Find 10% of 140.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 10/100 × 140 = 14.

4. State the result in the form requested: 14.

Answer: 14

Worked example 2

A quantity changes from a reference whole of 100 to a compared amount of 40. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (40 ÷ 100) × 100 = 40%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 40%; denominator 100

Day 102 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 25% of 200.

02 / Constructed response

10% of a number is 18. Find the number and check.

03 / Constructed response

10 is what percent of 40?

04 / Constructed response

Find 30% of 200.

05 / Constructed response

75% of a number is 150. Find the number and check.

06 / Constructed response

40 is what percent of 100?

07 / Constructed response

Find 20% of 60. Then explain a mental-math or fraction strategy that checks your answer.

08 / Multiple select

32 is 40% of an unknown quantity w. Select all equations that find w.
A. w = 32 + 40
B. w = 32 ÷ (40/100)
C. w = 32 × (40/100)
D. (40/100)w = 32

09 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 9. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

Find 125% of 180. Then explain a mental-math or fraction strategy that checks your answer.

Day 102 | Unit 07 | Answer key

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1. 50

25/100 × 200 = 50.

6.RP.A.3c | Constructed response

2. 180

Whole = 18 ÷ 1/10 = 180. Check 10% × 180 = 18.

6.RP.A.3c | Constructed response

3. 25%

10 ÷ 40 × 100 = 25%; the reference whole is 40.

6.RP.A.3c | Constructed response

4. 60

30/100 × 200 = 60.

6.RP.A.3c | Constructed response

5. 200

Whole = 150 ÷ 3/4 = 200. Check 75% × 200 = 150.

6.RP.A.3c | Constructed response

6. 40%

40 ÷ 100 × 100 = 40%; the reference whole is 100.

6.RP.A.3c | Constructed response

7. 12

20/100 × 60 = 12.

6.RP.A.3c | Constructed response

8. B, D - w = 32 ÷ (40/100); (40/100)w = 32

The part is percent rate times whole. Divide 32 by 2/5; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

9. 15%; denominator 60

Percent = (9 ÷ 60) × 100 = 15%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 225

125/100 × 180 = 225.

6.RP.A.3c | Constructed response